1. Proof the $(S^1, \times)$ is the group.

Note: $S^1 \triangleq \{z \in C \mid |z| = 1\}$

① 封闭性

$\forall z_1 \in S_1, z_2 \in S_2$, $|z_1|=1, |z_2|=1$

则有 $|z_1 z_2| = |z_1||z_2| = 1$, 所以 $z_1 z_2 \in S^1$

② 单位元

取 $e=1$, 由于 $|1|=1$, 所以 $1 \in S^1$.

$\forall z \in S^1$, $z \times 1 = z$, 所以 $e=1$ 是该群的单位元

③ 逆元

将 z 表示为复数形式, $z = e^{i\theta} \in S^1$

对于任意 $z = e^{i\theta} \in S^1$, 存在 $z^{-1} = e^{-i\theta}$

由于 $|e^{-i\theta}| = 1$, 所以 $z^{-1} \in S^1$

且 $z \times z^{-1} = e^{i\theta} e^{-i\theta} = 1 = e$, 每个元素都存在唯一逆元

④ 结合律: 复数乘法本身满足结合律

所以 $(S^1, \times)$ 是一个群

2. Given $R \in SO(3), v, \omega \in R^3$:

show that $R\hat{\omega}R^T = (R\omega)^\wedge$;

show that $R(v \times \omega) = (Rv \times R\omega)$.

(1) $R\hat{\omega}R^T = (R\omega)^\wedge$

① 确认 $S = R\hat{\omega}R^T$ 是反对称矩阵

$S^T = (R\hat{\omega}R^T)^T = (R^T)^T \hat{\omega}^T R^T = R\hat{\omega}^T R^T$

因为 $\hat{\omega}$ 是反对称矩阵, 所以 $\hat{\omega}^T = -\hat{\omega}$

所以 $S^T = -R\hat{\omega}R^T = -S$, 说明 S 也是一个 3×3 的反对称矩阵

那么 S 一定对应一个三维向量 p, 则 $S = \hat{p}$

根据反对称矩阵的结构 $S = \hat{p} = \begin{bmatrix} 0 & -p_3 & p_2 \\ p_3 & 0 & -p_1 \\ -p_2 & p_1 & 0 \end{bmatrix}$

$p_1 = S_{32}$, $p_2 = S_{13}$, $p_3 = S_{21}$

则当前目标就是算出 p_1, p_2, p_3, 看看是否等于向量 $R\omega$

② 用R的行向量计算 S_{ij}

$R=\begin{bmatrix} r_1^T \\ r_2^T \\ r_3^T \end{bmatrix}$ $\quad S_{ij}=r_i^T \hat{w} r_j = r_i^T (w \times r_j) = w(r_j \times r_i)$

③代入 SO(3)的性质

$p_1=S_{32}=w(r_2 \times r_3)=w \cdot r_1 = r_1^T w$

$p_2=S_{13}=w(r_3 \times r_1)= w \cdot r_2 = r_2^T w$

$p_3=S_{21}=w(r_1 \times r_2)= w \cdot r_3 = r_3^T w$

将三个分量叠着拼起来得到向量 p

$p=\begin{bmatrix} r_1^T w \\ r_2^T w \\ r_3^T w \end{bmatrix}=\begin{bmatrix} r_1^T \\ r_2^T \\ r_3^T \end{bmatrix} w = Rw$

因为 $S \to p$，且 $p=Rw$，综上，$R\hat{w}R^T=(Rw)^\wedge$

2、$R(v \times w)= Rv \times Rw$

$(Rv \times Rw)=(Rv)^\wedge Rw = R\hat{v}R^T Rw = R\hat{v}w = R(v \times w)$

3. A vector v is rotated about Z_A by θ degrees and is subsequently rotated about X_A by φ degrees. Give the rotation matrix R_{ab} that accomplished these rotations in the given order. If $\theta = 30$ degrees and $\varphi = 45$ degrees, calculate the rotation matrix.

$R_{ab} = R_x(\varphi) R_z(\theta)$

$R_x(\varphi)= e^{\hat{x}\varphi}=\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix}$

$R_y(\theta)=e^{\hat{y}\theta} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$

$R_z(\psi)=e^{\hat{z}\psi}=\begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[illegible] $R_z(\psi)= e^{\hat{z}\psi}$，

$z=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $\quad \hat{z}=\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Rodrigue's rotation Formula:

$e^{\hat{w}\theta}=I+\hat{w}\sin\theta+\hat{w}^2(1-\cos\theta)$

$$e^{\hat{k}\psi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sin\psi + \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}(1-\cos\psi)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -\sin\psi & 0 \\ \sin\psi & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} \cos\psi - 1 & 0 & 0 \\ 0 & \cos\psi - 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{ab} = R_x(\varphi) \cdot R_z(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \cos\varphi\sin\theta & \cos\varphi\cos\theta & -\sin\varphi \\ \sin\varphi\sin\theta & \sin\varphi\cos\theta & \cos\varphi \end{bmatrix}$$

当 $\theta = 30°$, $\varphi = 45°$ 时 $R_{ab} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{6}}{4} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{4} & \frac{\sqrt{6}}{4} & \frac{\sqrt{2}}{2} \end{bmatrix}$

4. A frame $\{B\}$ is located initially coincident with a frame $\{A\}$. We rotate $\{B\}$ about Z_B by θ degrees, and then we rotate the resulting frame about X_B by φ degrees. Find the rotation matrix R_{ab} that will change the description of vectors from $\{B\}$ to $\{A\}$. If $\theta = 30$ degrees and $\varphi = 45$ degrees, calculate the rotation matrix.

欧拉角 旋转矩阵 右乘

$$R_{ab} = R_z(\theta) R_x(\varphi)$$

$$R_{ab} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & -\sin\varphi \\ 0 & \sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} c\theta & -s\theta c\varphi & c\varphi \\ s\theta & c\theta c\varphi & -c\varphi s\varphi \\ 0 & s\varphi & c\varphi \end{bmatrix}$$

当 $\theta = 30°$, $\varphi = 45°$ 时 $R_{ab} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}}{2} \\ \frac{1}{2} & \frac{\sqrt{6}}{4} & -\frac{\sqrt{6}}{4} \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$

Homework 2 **5. Matlab programing**

a) Write a Matlab program function that transforms rigid transformation matrices fromthe **ZYZ convention** to the **exponential** representation, named **zyz2exp**; and write aMatlab program function that transforms rigid transformation matrices from the **exponential representation** to the **ZYZ** convention, named **exp2zyz**. Submit your Matlab code and give two examples to verify your program.

b) Write a Matlab program function that transforms rigid transformation matrices from the **RPY convention** to the **exponential** representation, named **rpy2exp**; and write a program function that transforms rigid transformation matrices from the **exponential representation** to the **RPY** convention, named **exp2rpy**. Submit your Matlab code and give two examples to verify your program.

Matlab 代码

```
% ========================================================================
% 机器人学 Homework 2 - 刚体变换的参数化转换
% 包含: zyz2exp, exp2zyz, rpy2exp, exp2rpy 以及测试用例
% ========================================================================

function hw2_kinematics()
    clc; clear;

    disp('=========== 测试用例 1: ZYZ 约定与指数表示的转换 ===========');
    % 设定一个初始的 ZYZ 角度 (弧度) 和平移向量
    zyz_in = [pi/4, pi/3, pi/6];
    p_in = [10; 20; 30];

    % 1. ZYZ 转换为 指数表示
    [xi, theta] = zyz2exp(zyz_in, p_in);
    disp('ZYZ 转 Exp 结果:');
    fprintf('Theta = %.4f rad\n', theta);
    disp('Twist Xi [v; omega] = '); disp(xi);

    % 2. 指数表示转回 ZYZ
    [zyz_out, p_out] = exp2zyz(xi, theta);
    disp('Exp 转回 ZYZ 结果 (应与输入一致):');
    disp('ZYZ Angles = '); disp(zyz_out);
    disp('Translation p = '); disp(p_out);

    disp('=========== 测试用例 2: RPY 约定与指数表示的转换 ===========');
    % 设定一个初始的 RPY 角度 (Roll, Pitch, Yaw) 和平移向量
    rpy_in = [pi/6, pi/4, pi/3];
    p_in2 = [-5; 15; 8];

    % 1. RPY 转换为 指数表示
    [xi2, theta2] = rpy2exp(rpy_in, p_in2);
    disp('RPY 转 Exp 结果:');
    fprintf('Theta = %.4f rad\n', theta2);
    disp('Twist Xi [v; omega] = '); disp(xi2);

    % 2. 指数表示转回 RPY
    [rpy_out, p_out2] = exp2rpy(xi2, theta2);
    disp('Exp 转回 RPY 结果 (应与输入一致):');
    disp('RPY Angles = '); disp(rpy_out);
    disp('Translation p = '); disp(p_out2);
end

% ========================================================================
% 核心函数 1: ZYZ 到 指数映射
function [xi, theta] = zyz2exp(zyz, p)
    a = zyz(1); b = zyz(2); c = zyz(3);
    % 构造旋转矩阵 Rz(a)*Ry(b)*Rz(c)
    Rz1 = [cos(a) -sin(a) 0; sin(a) cos(a) 0; 0 0 1];
```

```
    Ry  = [cos(b) 0 sin(b); 0 1 0; -sin(b) 0 cos(b)];
    Rz2 = [cos(c) -sin(c) 0; sin(c) cos(c) 0; 0 0 1];
    R = Rz1 * Ry * Rz2;

    [xi, theta] = matrixLog(R, p);
end

% 核心函数 2: 指数映射 到 ZYZ
function [zyz, p] = exp2zyz(xi, theta)
    [R, p] = matrixExp(xi, theta);
    % 提取 ZYZ 欧拉角
    beta = atan2(sqrt(R(3,1)^2 + R(3,2)^2), R(3,3));
    if abs(sin(beta)) < 1e-6  % 奇异点(万向节死锁)处理
        alpha = 0;
        gamma = atan2(-R(1,2), R(1,1));
    else
        alpha = atan2(R(2,3), R(1,3));
        gamma = atan2(R(3,2), -R(3,1));
    end
    zyz = [alpha, beta, gamma];
end

% 核心函数 3: RPY 到 指数映射
function [xi, theta] = rpy2exp(rpy, p)
    r = rpy(1); pt = rpy(2); y = rpy(3);
    % 构造旋转矩阵 Rz(y)*Ry(pt)*Rx(r)
    Rx = [1 0 0; 0 cos(r) -sin(r); 0 sin(r) cos(r)];
    Ry = [cos(pt) 0 sin(pt); 0 1 0; -sin(pt) 0 cos(pt)];
    Rz = [cos(y) -sin(y) 0; sin(y) cos(y) 0; 0 0 1];
    R = Rz * Ry * Rx;

    [xi, theta] = matrixLog(R, p);
end

% 核心函数 4: 指数映射 到 RPY
function [rpy, p] = exp2rpy(xi, theta)
    [R, p] = matrixExp(xi, theta);
    % 提取 RPY 角
    pitch = atan2(-R(3,1), sqrt(R(3,2)^2 + R(3,3)^2));
    if abs(cos(pitch)) < 1e-6 % 奇异点(万向节死锁)处理
        roll = 0;
        yaw = atan2(R(2,1), R(2,2));
    else
        roll = atan2(R(3,2), R(3,3));
        yaw = atan2(R(2,1), R(1,1));
    end
    rpy = [roll, pitch, yaw];
end

% ========================================================================
% 底层数学工具库 (李群与李代数转换)

% 矩阵对数 (SO3/SE3 -> 旋量)
function [xi, theta] = matrixLog(R, p)
    theta = acos((trace(R) - 1) / 2);
    if abs(theta) < 1e-6
        omega = [0; 0; 0];
        theta = norm(p);
        if theta < 1e-6
            v = [0; 0; 0];
        else
            v = p / theta;
        end
    else
        omega_hat = (R - R') / (2 * sin(theta));
```

```
        omega = [omega_hat(3,2); omega_hat(1,3); omega_hat(2,1)];
        % 计算平移映射矩阵 G
        G = eye(3)*theta + (1 - cos(theta))*omega_hat + (theta - sin(theta))*omega_hat^2;
        v = G \ p; % 等价于 inv(G)*p，提取线速度分量
    end
    xi = [v; omega];
end

% 指数映射 (旋量 -> SO3/SE3)
function [R, p] = matrixExp(xi, theta)
    v = xi(1:3); omega = xi(4:6);
    omega_hat = [0 -omega(3) omega(2); omega(3) 0 -omega(1); -omega(2) omega(1) 0];

    if norm(omega) < 1e-6
        R = eye(3);
        p = v * theta;
    else
        % 罗德里格斯公式计算旋转矩阵 R
        R = eye(3) + omega_hat * sin(theta) + omega_hat^2 * (1 - cos(theta));
        % 计算平移向量 p
        G = eye(3)*theta + (1 - cos(theta))*omega_hat + (theta - sin(theta))*omega_hat^2;
        p = G * v;
    end
end
```

输出结果

```
=========== 测试用例 1: ZYZ 约定与指数表示的转换 ===========
ZYZ 转 Exp 结果:
Theta = 1.6267 rad
Twist Xi [v; omega] =
    1.3099
    7.8231
   22.0479
   -0.0898
    0.6823
    0.7256

Exp 转回 ZYZ 结果 (应与输入一致):
ZYZ Angles =
    0.7854    1.0472    0.5236

Translation p =
    10
    20
    30

=========== 测试用例 2: RPY 约定与指数表示的转换 ===========
RPY 转 Exp 结果:
Theta = 1.2105 rad
Twist Xi [v; omega] =
   -1.8868
   13.8963
    4.6348
    0.0391
    0.7728
    0.6335

Exp 转回 RPY 结果 (应与输入一致):
RPY Angles =
    0.5236    0.7854    1.0472
```

```
Translation p =
   -5.0000
   15.0000
    8.0000
```